

Katakko: Embodiment of Modular Robots through Automatic Motion Mapping

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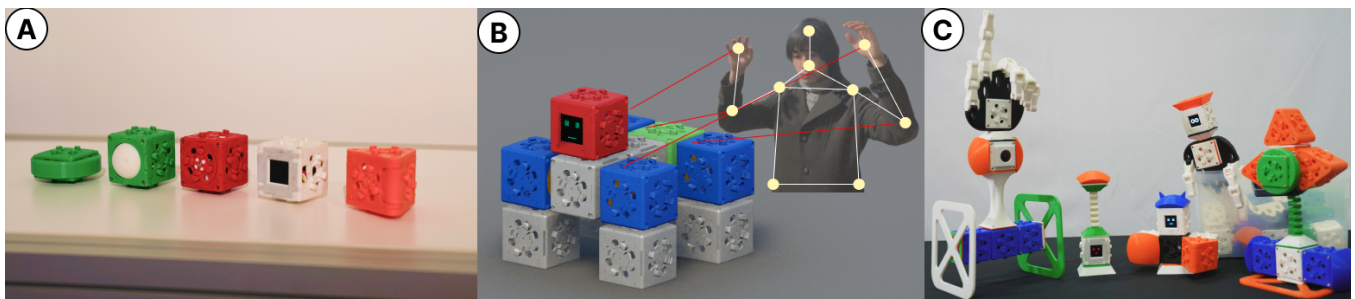


Figure 1: (A) Communication-oriented modules designed for social interaction. (B) A robot assembled by the user from Katakko modules. The system automatically generates joint-to-joint mappings between the user's upper body and the assembled robot's joint structure. (C) The user teleoperates the robot in real time, with the robot mirroring the user's body movements through the generated mappings.

Abstract

This paper presents Katakko, a modular robot system that enables users to create and embody personalized social agents. While modular robots offer high design flexibility, their potential for social communication remains largely untapped; existing platforms often prioritize basic functional tasks, lacking both the communication-oriented hardware and the intuitive control mapping essential for social interaction. Katakko addresses these limitations through two core contributions: a suite of four module types (display, actuator, mobile, and sensor) and an optimization-based motion mapping algorithm. Our framework automatically extracts the hierarchical structure of a physical assembly via video-based detection and generates joint-to-joint mappings by aligning human and robot hierarchies. Furthermore, the system supports incremental updates, preserving the user's established control intuition even as the robot's form evolves through reconfiguration.

*These authors contributed equally to this research.

CCS Concepts

• **Human-centered computing** → **Interaction design**; • **Computer systems organization** → **Robotics**.

Keywords

Modular robotics, telepresence, motion mapping, tangible interface

ACM Reference Format:

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1 Introduction

Modular robots are composed of reconfigurable functional units that can be rearranged to form different body structures, enabling a single system to adapt to diverse tasks such as search-and-rescue, environmental monitoring [Fukuda and Kawauchi 1990; Sproewitz et al. 2009]. This reconfigurable capability is being researched as a tangible interface in the field of human-computer interaction [Raffle et al. 2004]. However, a promising yet underexplored application is social communication. Because users can freely assemble modules into a personalized body structure, including non-anthropomorphic features such as animal ears, tails, or additional limbs, modular



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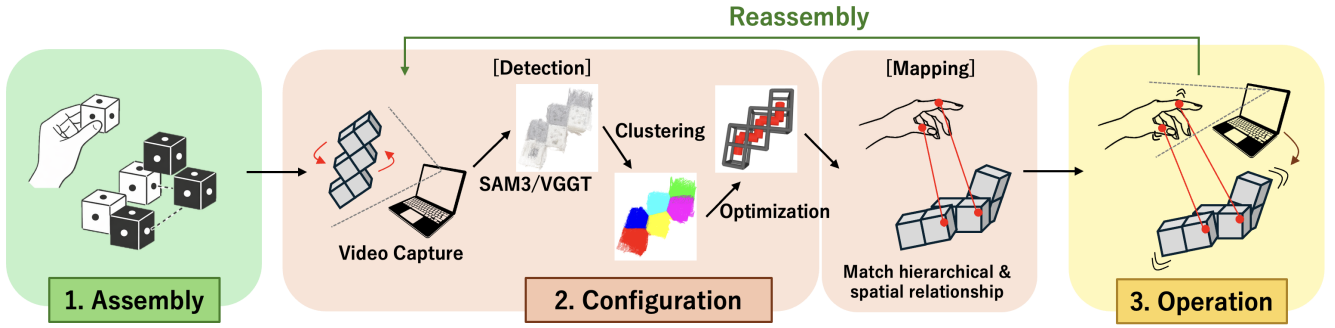


Figure 2: System overview: Katakko enables an iterative loop of assembly, configuration, and operation. Users first assemble cube modules into a custom robot. The system then detects the robot’s structure via video-based point-cloud reconstruction (SAM3/VGGT) and automatically generates joint-to-joint mappings between the user’s body and the robot. Users teleoperate the robot in real time, and upon reassembly, the system incrementally updates the mapping to preserve the previous control scheme.

robots have the potential to serve as customizable social agents that reflect the user’s identity and preferences.

Realizing this vision presents two critical challenges that existing platforms do not adequately address. First, most current systems are designed primarily for locomotion or manipulation, lacking communication-oriented modules such as gaze displays, expressive actuators, and audio-visual sensing, necessary for social interaction. Second, when a user creates a robot with an arbitrary, non-anthropomorphic topology, there is no established method to automatically generate a motion mapping between the user’s body and the robot’s configuration. Requiring manual definition for every structural change severely limits the iterative design loop and disrupts the sense of social embodiment.

To bridge this gap, we present *Katakko*, a modular robot system designed for user-customizable social communication. Katakko contributes communication-oriented modules in four categories, display, actuator, mobile, and sensor, that users can assemble into a personalized agent. To enable intuitive control of arbitrary structures, we introduce a joint-wise mapping algorithm that aligns the parent–child relationships and spatial arrangement of human and robot joints. The system also supports incremental updates, preserving existing mappings while adapting to added or removed joints, enabling a smooth loop of assembly, configuration, and operation, thereby maintaining a consistent “social identity” across different forms.

2 System Design

Our system consists of four components as shown in fig. 2: (1) cube robot modules equipped with actuators and displays to achieve social interaction, (2) a video-based detector that estimates the structure of the assembled robot, (3) a joint-wise mapping generator that aligns human and robot joints, and (4) an interface that tracks user motion and streams control commands to the robot.

All modules are built from off-the-shelf components and 3D-printed parts for accessibility. A well-designed side latch mechanism allows cubes to be freely connected without screws or embedded magnets. The video-based detector infers the spatial relationships

between cubes as the user rotates the robot in front of a camera by clustering cube point-cloud representations reconstructed with SAM3 [Carion et al. 2025] and VGGT [Wang et al. 2025], eliminating the need for tedious manual modeling. Inspired by HandAvatar [Jiang et al. 2023], the mapping generator produces joint matching that consider the joint hierarchies and spatial arrangement of both the human and the robot, enabling intuitive control. Furthermore, after minor reassembly, the system detects structural changes and regenerates a mapping that preserves the previous one as much as possible, enabling a continuous loop of assembly, configuration, and operation.

3 Demonstration

Visitors experience the full workflow of Katakko through three steps (fig. 2):

- **Assembly:** Visitors assemble communication-oriented modules into a personalized robot using mechanical connectors, freely exploring both anthropomorphic and non-anthropomorphic configurations.
- **Configuration:** The system automatically generates joint-to-joint mappings from the visitor’s full-body motion to the assembled robot’s joint configuration.
- **Operation:** Visitors teleoperate the robot in real time using their body movements. They can then reconfigure the modules and experience how the system incrementally adapts the control mapping to the new design.

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References

- Nicolas Carion, Laura Gustafson, Yuan-Ting Hu, Shoubhik Debnath, Ronghang Hu, Dac Suris, Chaitanya Ryali, Kalyan Vasudev Alwala, Haitham Khedr, Andrew Huang, et al. 2025. Sam 3: Segment anything with concepts. *arXiv preprint arXiv:2511.16719* (2025).
- T. Fukuda and Y. Kawauchi. 1990. Cellular robotic system (CEBOT) as one of the realization of self-organizing intelligent universal manipulator. In *Proceedings, IEEE International Conference on Robotics and Automation*. 662–667 vol.1. doi:10.1109/ROBOT.1990.126059
- Yu Jiang, Zhipeng Li, Mufei He, David Lindlbauer, and Yukang Yan. 2023. Handavatar: Embodying non-humanoid virtual avatars through hands. In *Proceedings of the 2023 CHI conference on human factors in computing systems*. 1–17.
- Hayes Solos Raffle, Amanda J. Parkes, and Hiroshi Ishii. 2004. Topobo: a constructive assembly system with kinetic memory. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Vienna, Austria) (*CHI '04*). Association for Computing Machinery, New York, NY, USA, 647–654. doi:10.1145/985692.985774
- Alexander Sproewitz, Aude Billard, Pierre Dillenbourg, and Auke Jan Ijspeert. 2009. Roombots-mechanical design of self-reconfiguring modular robots for adaptive furniture. In *2009 IEEE International Conference on Robotics and Automation*. 4259–4264. doi:10.1109/ROBOT.2009.5152613
- Jianyuan Wang, Minghao Chen, Nikita Karaev, Andrea Vedaldi, Christian Rupprecht, and David Novotny. 2025. Vgggt: Visual geometry grounded transformer. In *Proceedings of the Computer Vision and Pattern Recognition Conference*. 5294–5306.